

Affinoid spaces

Fix a non-archimedean completed valued field $(\mathbf{k}, |\cdot|)$ throughout this section. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. It should be viewed as the non-archimedean analogue of an affine variety in algebraic geometry or an affine scheme in scheme theory. The goal of this section is to define a suitable Grothendieck topology on X such that we can define a structure sheaf of analytic functions on X , and then study its basic properties.

1 Affinoid domains

On a general open subset $U \subseteq X$, the ring of analytic functions on U may behave badly.

Yang: To be continued.

Definition 1. A closed subset $V \subseteq X$ is called an *affinoid domain* if there exists a $\mathbf{k}$ -affinoid algebra A_V and a morphism of $\mathbf{k}$ -affinoid algebras $\theta : A \rightarrow A_V$ satisfying the following universal property: for every bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\psi : A \rightarrow B$ such that the induced map on spectra $\mathcal{M}(\psi) : \mathcal{M}(B) \rightarrow X$ has its image contained in V , there exists a unique bounded homomorphism $\varphi : A_V \rightarrow B$ such that the following right diagram commutes:

$$\begin{array}{ccc} & V & \\ \swarrow & & \nwarrow \\ X & \xleftarrow{\mathcal{M}(\psi)} & \mathcal{M}(B) \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} & A_V & \\ \theta \nearrow & & \searrow \varphi \\ A & \xrightarrow{\psi} & B \end{array}$$

In this case, we say that V is represented by the $\mathbf{k}$ -affinoid algebra A_V . If A_V is strict, then we say that V is a strict affinoid domain.

Slogan A closed subset $V \subset X$ is an affinoid domain if the functor “ $\text{Mor}(-, V)$ ” is representable by a $\mathbf{k}$ -affinoid algebra A_V .

There are several important classes of affinoid domains.

Construction 2. Let $f = (f_1, \dots, f_n), g = (g_1, \dots, g_m)$ be two tuples of elements in A and $r = (r_1, \dots, r_n), s = (s_1, \dots, s_m)$ be two tuples of positive real numbers. Consider the following closed subset of X :

$$X(\underline{f/r}; \underline{s/g}) := \{x \in X : |f_i(x)| \leq r_i, |g_j(x)| \geq s_j, 1 \leq i \leq n, 1 \leq j \leq m\}.$$

Such a closed subset is called a *Laurent domain* of X . If $m = 0$, then it is called a *Weierstrass domain* of X , denoted by

$$X(\underline{f/r}) := \{x \in X : |f_i(x)| \leq r_i, 1 \leq i \leq n\}.$$

More generally, let $f = (f_1, \dots, f_n), g$ be elements in A such that the ideal generated by them is the whole algebra A . Set $p = (p_1, \dots, p_n)$ be a tuple of positive real numbers. We define the following closed subset of X :

$$X(\underline{(f/g)/p}) := \{x \in X : |f_i(x)| \leq p_i |g(x)|, 1 \leq i \leq n\}.$$

Such a closed subset is called a *rational domain* of X .

Proposition 3. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Then any Weierstrass domain, Laurent domain, or rational domain of X is an affinoid domain. Moreover, if A is strict, then these affinoid domains are all strict. *Yang: To be checked.*

Proof. *Yang: To be added.* □

We have a sequence of inclusion:

$$\{\text{Affinoid domains}\} \supseteq \{\text{Rational domains}\} \supseteq \{\text{Laurent domains}\} \supseteq \{\text{Weierstrass domains}\}.$$

Yang: Why Laurent domain is rational domain?

Proposition 4. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural map on spectra $\mathcal{M}(\theta) : \mathcal{M}(A_V) \rightarrow X$ induced by the representing homomorphism $\theta : A \rightarrow A_V$ has its image equal to V , and induces a homeomorphism $\mathcal{M}(A_V) \cong V$.

Proof. *Yang: To be added.* □

Proposition 5. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural bounded homomorphism $\phi : A \rightarrow A_V$ is flat.

Proof. *Yang: To be added.* □

Proposition 6. Let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the inverse image $f^{-1}(V) \subseteq Y$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $B \hat{\otimes}_A A_V$. *Yang: To be revised.*

Proof. *Yang: To be added.* □

Proposition 7. Let $V_1, V_2 \subseteq X$ be two affinoid domains represented by the $\mathbf{k}$ -affinoid algebras A_{V_1}, A_{V_2} , respectively. Then the intersection $V_1 \cap V_2 \subseteq X$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $A_{V_1} \hat{\otimes}_A A_{V_2}$. *Yang: To be checked.*

Proof. *Yang: To be added.* □

Proposition 8. Let $x \in X$ be a point. Then the collection of strict affinoid domains of X containing x forms a fundamental system of neighborhoods of x . Explicitly, for any open neighborhood $U \subseteq X$ of x , there exists a strict affinoid domain $V \subseteq X$ such that $x \in V \subseteq U$. *Yang: To be checked.*

Theorem 9 (Gerritzen-Grauert theorem). Let $V \subseteq X$ be an affinoid domain. Then there exist finitely many rational domains $V_1, \dots, V_m \subseteq X$ such that $V = V_1 \cup \dots \cup V_m$. *Yang: To be checked.*

Proof. *Yang: To be added.* □

Definition 10. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Suppose that $A_V \hat{\otimes} \mathbf{K}$ is strictly $\mathbf{K}$ -affinoid for some completed valued field extension $\mathbf{K}/\mathbf{k}$. The *dimension* of V is defined as the Krull dimension of the affinoid algebra $A_V \hat{\otimes} \mathbf{K}$:

$$\dim(V) := \dim_{\text{Krull}}(A_V \hat{\otimes} \mathbf{K}).$$

By [Proposition 15](#), the definition is independent of the choice of $\mathbf{K}$.

2 The Grothendieck topology and structure sheaf

Definition 11. A *special domain* of X is a finite union of affinoid domains of X .

Definition 12. The *site of affinoid domains* on X , is defined as follows:

- the underlying category has as objects the affinoid domains $V \subseteq X$, and the morphisms are given by the inclusion maps;
- a covering of an objects $V \subseteq X$ is a finite family of affinoid domains $\{V_i\}_{i \in I}$ such that $V = \bigcup_{i \in I} V_i$.

Yang: To be revised.

Definition 13. For every open subset $U \subseteq X$, we define the ring of analytic functions on U as follows:

$$\mathcal{O}_X(U) := \varprojlim_{V \subseteq U} A_V,$$

where the limit is taken over all affinoid domains $V \subseteq U$, and A_V is the $\mathbf{k}$ -affinoid algebra representing the affinoid domain V .

Theorem 14 (Acyclicity theorem). Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $\{V_i\}_{i \in I}$ be a finite covering of X by affinoid domains. For each finite subset $J \subseteq I$, set $V_J := \bigcap_{j \in J} V_j$, which is an affinoid domain by [Proposition 7](#). Then the augmented Čech complex

$$0 \rightarrow A \rightarrow \prod_{i \in I} A_{V_i} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=2}} A_{V_J} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=3}} A_{V_J} \rightarrow \dots$$

is exact. Yang: To be checked.

3 Relative boundary and interior

Appendix

Proposition 15. Let A be a strictly $\mathbf{k}$ -affinoid algebra. For any completed valued field extension $\mathbf{K}/\mathbf{k}$, we have

$$\dim_{\mathbf{K}\text{rull}}(A) = \dim_{\mathbf{K}\text{rull}}(A \hat{\otimes}_{\mathbf{k}} \mathbf{K}).$$

Yang: To be checked.